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RESEARCH ARTICLE

Cheby-KANs: Advanced Kolmogorov–Arnold Networks for Applying Geometric Deep Learning in Quantum Chemistry Applications

AMGAD ABDALLAH MAHMOUD¹, ANDREAS PESTER¹, (Member, IEEE),
MAHA M. MUTTARDI², FREDERIC ANDRES³, (Senior Member, IEEE), SHIHORI TANABE⁴,
NICOLAS GRENECHE⁵, AND HESHAM H. ALI⁶

¹AI Group, Faculty of Informatics and Computer Science, The British University in Egypt, Al Shorouk City, Cairo 11837, Egypt

²Mathematics and Actuarial Science Department, American University in Cairo, New Cairo City 11835, Egypt

³National Institute of Informatics, Chiyoda, Tokyo 101-8430, Japan

⁴Division of Risk Assessment, Center for Biological Safety and Research, National Institute of Health Sciences, Kawasaki 210-9501, Japan

⁵LIPN, UMR 7030, University of Sorbonne Paris North, 93430 Villetaneuse, France

⁶College of Information Science and Technology, University of Nebraska at Omaha, Omaha, NE 68182, USA

Corresponding author: Amgad Abdallah Mahmoud (amgad.abdallah@bue.edu.eg)

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ABSTRACT In this work, we present an enhanced version of the Kolmogorov–Arnold network (KAN) algorithm, called Cheby-KAN, that offers a more efficient, more reliable, and faster alternative to conventional KANs. We integrated our algorithm with a geometric deep learning model (SchNet) to assess the effect of integrating KANs representation on geometric deep learning by combining domain knowledge of quantum chemistry and our new representation. We initially tested our model on benchmark datasets, then we integrated Cheby-KAN with SchNet to obtain an enhanced model, called Cheby-KAN-SchNet. We used Cheby-KAN-SchNet to predict six quantum properties of molecules, and compared our results against the original SchNet results and another model integrating SchNet with the original KANs using basis splines to give a fair and objective comparison of our model. We expected to obtain the benefits of the powerful KANs representation and to reduce the high time and computational costs of KANs. We applied the algorithm to quantum chemistry because quantum chemistry requires extremely high precision modeling of the system where error rates increase exponentially. Our algorithm outperformed the original KANs with B-splines in terms of both speed and accuracy, and outperformed Schnet in terms of accuracy and consistency. These results demonstrate the potential of Cheby-KAN in addressing the approximation of multi-variate and complex functions under high levels of uncertainty, while offering enhanced interpretability compared to traditional neural network models or KANs with B-splines.

INDEX TERMS Geometric deep learning, Kolmogorov–Arnold networks, neural networks representation, quantum chemistry.

I. INTRODUCTION

Advances in artificial intelligence (AI) over the past few decades have been remarkable, with multi-layer perceptrons (MLPs) being used as an essential element, either as a core building block for various applications as a standalone representation or inside another deep learning

(DL) architecture. MLPs are widely used because they can, in theory, approximate any continuous and differentiable function given a finite number of neurons, several layers, and chosen non-linear activation functions. However, there is a caveat because these activation functions are predefined and fixed before training, which can limit the model's expressiveness and make it hard to understand how decisions are made or what the original function was that describes the data, which is called low interoperability.

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In 1957, Vladimir Arnold, a student of the legendary mathematician Andrey Kolmogorov, introduced a groundbreaking concept. He showed that any continuous multivariate function, $f : [0, 1]^n \rightarrow \mathbb{R}$, could be constructed as a combination of simpler, one-dimensional functions, Φ_q and $\psi_{q,p}$. This is known as the Kolmogorov–Arnold representation theorem [1], which can be described using the expression

$$f(x_1, \dots, x_n) = \sum_{q=0}^{2n} \Phi_q \left(\sum_{p=1}^n \psi_{q,p}(x_p) \right) \quad (1)$$

where

- $\psi_{(q,p)} : [0, 1] \rightarrow \mathbb{R}$
- $\Phi_q : \mathbb{R} \rightarrow \mathbb{R}$

In 2024, Liu et al. [2] used this representation to develop an alternative to MLPs, which have been the core building block of almost all AI applications, a feedforward algorithm over more than one hidden layer with backpropagation for training. MLPs, based on the universal approximation theorem, can in theory, approximate any real-valued continuous function, $M : \mathbb{R}^n \rightarrow \mathbb{R}$, by a neural network with a finite number of parameters and even one layer, that is,

$$M(x_1, \dots, x_n) = \sum_{i=1}^N \alpha_i \phi_R \left(\sum_{j=1}^K w_{ij} x_j + b_i \right) \quad (2)$$

where

- α_i is the weight associated with the output of each neuron,
- ϕ_R is a fixed activation function predefined in the model by the user,
- w_i and b_i are the weight and bias terms, respectively.

However, MLPs rely on a set of activation functions that are predefined before training the neural network and are applied consistently to every node. These activation functions are part of the MLP hyperparameters, which are chosen and fixed before training. The network's expressiveness may be limited by this rigidity, which makes it difficult to learn complex correlations between inputs and outputs. This approach also limits the explainability of MLPs, which are often called black boxes. In contrast, Kolmogorov–Arnold networks (KANs) learn activation functions on the edges of the network, and these functions are subsequently used to process the input data. This gives KANs greater flexibility, the capacity to represent complex functions, and improved interpretability by allowing every weight parameter to be replaced with a univariate function that is specified in the original version as a basis spline (B-spline) [3]. Unlike MLPs, which incorporate linear transformations followed by nonlinear activation functions, each node in a KAN adds up these function outputs without applying any nonlinear transformations. The Kolmogorov–Arnold representation theorem, which explains how continuous functions can be represented by a finite number of operations, inspired this

innovative architecture (see (3) and (4)).

$$\text{MLP}(x) = x (W_0 \circ \sigma \circ W_1 \circ \dots \circ \sigma \circ W_L) \quad (3)$$

$$\text{KAN}(x) = x (\Phi_0 \circ \Phi_1 \circ \dots \circ \Phi_L) \quad (4)$$

It has been claimed that KANs theoretically have a parameter efficiency 100 times greater than that of MLPs [2], which can be beneficial in training large models with fewer parameters and is an advantage of KANs over MLPs. This benefit is also key in scalability. Nevertheless, this prediction is theoretical and has only been assessed with a few examples on small datasets. In this paper, we integrate KANs with SchNet [4], which already has a track record of use, to predict six quantum properties. MLPs are estimated to be 10 times faster than KANs in CPUs and GPUs struggle with KANs.

In 1989, Girosi and Poggio applied the Kolmogorov–Arnold theorem to artificial neural networks and they found it irrelevant in the context of networks for learning [5]. Liu et al. [6] applied several KAN layers (deeper network) and overcame the limitations shown by Girosi and Poggio to build an effective algorithmic structure.

II. BACKGROUND

A. B-SPLINES

In KANs, B-splines represent the core feature because they help the network to learn and adapt in a highly controlled manner; B-splines allow KANs to memorize what they have learned by keeping changes localized. B-spline functions used in KANs are linear combinations of mostly cubic polynomials [7]. The spline function parameterizes a generic one-dimensional curve, like other interpolation polynomials, but it avoids the problematic behavior of high-degree polynomials (e.g., Bernstein polynomials [8]), such as oscillations at borders (Runge's phenomenon [9]). In general, a spline function of order n is constructed by dividing the domain into $k-1$ intervals, where the polynomial degree is $n-1$. These intervals are marked by a grid of k points on the x -axis, known as knots. Knots can be formulated as $t_0, \dots, t_k$. Within each interval $[t_i, t_{i+1}]$, the spline is defined by a polynomial of order n at most, ensuring that the overall function remains continuous, differentiable, and smooth at the knots. Cubic B-splines are used for their practical properties because they can be represented by a linear combination of B-spline basis functions, $B_{i,k}$. In terms of neural network computations, the choice of basis function is crucial, and we chose B-splines [10], [11] because they are usually well-conditioned. A B-spline can be formulated as

$$f(x) = \sum_{i=0}^n c_i B_{i,k}(x) \quad (5)$$

where c_i is the coefficient for the B-spline basis function of degree k .

For higher orders n , the B-splines are calculated recursively using the Cox–de Boor formula [12]. The functions for order n involve linear combinations of lower-order splines, defined

by

$$t_{i,n}^+(x) = \frac{x - t_i}{t_{i+n} - t_i} \quad (6)$$

and

$$t_{i,n}^-(x) = \frac{t_{i+n+1} - x}{t_{i+n+1} - t_{i+1}} \quad (7)$$

Then, the recursive relationship is given by

$$B_{i,n}(x) = t_{i,n}^+(x)B_{i,n-1}(x) + t_{i,n}^-(x)B_{i+1,n-1}(x) \quad (8)$$

One advantage of B-splines is the local control property, which means that each B-spline basis function is nonzero only within a limited interval, which is the spline support. This property allows the changes to the control points, or the learnable coefficients in case of KANs, of the spline to only affect the output in a localized region of the input space, adjusting the model without significantly altering the behavior of the function outside that region.

B. CHEBYSHEV POLYNOMIALS

Chebyshev polynomials of the first kind, which form the heart of our enhanced model, are a set of orthogonal polynomials. In our implementation, we used shifted Chebyshev polynomials, which are defined for the interval, $x \in [0, 1]$, rather than using the original Chebyshev polynomial, which is defined for the interval, $x \in [-1, 1]$, for the following main reasons.

- To match the Kolmogorov–Arnold representation presented in (1).
- To allow our model to handle applications on non-symmetric domains.

In this article, we refer to the “shifted Chebyshev polynomial” as the “Chebyshev polynomial” for brevity. For the n^{th} polynomial degree for each time step t , the Chebyshev polynomial can be defined by the analytical form [9]

$$T_n(x) = \cos(n \cdot \arccos(2x - 1)) \quad (9)$$

or with the recursive formula

$$T_0(x) = 1, \quad (10)$$

$$T_1(x) = 2x - 1, \quad (11)$$

$$T_{n+1}(x) = 2(x - 1)T_n(x) - T_{n-1}(x). \quad (12)$$

In our work, we used Chebyshev polynomials as the core method to approximate the nonlinear function on the edges of KANs, which resembles the activation functions in MLP. This is because Chebyshev polynomials have desirable properties, such as fast convergence [13] and orthogonality, that avoid some problems with numerical stability. We use the fifth degree Chebyshev polynomial, resulting in six bases. Moreover, Chebyshev polynomials help to minimize the Runge phenomenon, which is the large oscillations observed near the edges of the interval when using high-degree polynomials. This property is particularly valuable in our neural network model because it improves the stability and accuracy of function approximation across layers, which is useful when calculating integrals of activation-like functions in the KANs.

C. SchNet ARCHITECTURE

SchNet was originally designed for predicting molecular properties. It uses a continuous-filter convolutional (CFCnv) neural networks, and operates directly on raw molecular coordinates, making it suitable for tasks in computational chemistry and materials science. Initially, SchNet proposes a learnable embedding tensor of rank 2 $A : Z_i \mapsto X_i^0$, where Z_i is atomic number and X_i^0 is the initial atomic feature vector. Then, SchNet uses a continuous-filter convolutional operation to update atomic embeddings based on their neighbors, for which the updated embeddings for layer $l+1$ in a CFCnv layer for atom i is given by

$$x_i^{l+1} = \sigma \left(x_i^l + \sum_{j \neq i} W(e_{ij}) \odot x_j^l \right) \quad (13)$$

where

- σ is a fixed activation function (e.g., ReLU),
- $\odot$ is the element-wise multiplication.
- $e_{i,j} = \zeta(d_{i,j})$, where $d_{i,j}$ is the pairwise distance between atoms i and j .
- Finally, $d_{i,j} = \|r_i - r_j\|_2$, and r_i is the position of atom i .

Putting this all together gives us the formulation for a single SchNet layer as

$$x_i^{l+1} = x_i^l + \sum_{j \neq i} \text{MLP}(\zeta(d_{i,j})) \odot x_j^l \quad (14)$$

III. RELATED WORKS

KANs are relatively new representations, yet many researchers have used this method in different tasks to assess their strength and to compare them carefully with MLPs. Sulaiman et al. [14] compared the effectiveness of KANs, MLPs, and a hybrid teaching–learning-based optimization–deep learning (TLBO-DL) approach in predicting chiller energy consumption in commercial buildings. KANs showed superior predictive accuracy with the lowest RMSE and highest R^2 and their performance remained robust across various scenarios. The study showed that KANs are promising for more AI applications; the KANs outperformed MLPs and TLBO-DL methods in both precision and variance. Morwani et al. [15] explored a hybrid forecasting approach combining KANs and liquid neural networks (LNNs) for smart grid energy prediction. The architecture integrated KANs for initial function approximation, providing a strong foundation for accurate predictions, and it used LNNs for dynamic real-time adjustments based on temporal data. The hybrid architecture effectively balances accuracy, adaptability, and temporal sensitivity, making it a promising tool for real-world predictive tasks. Despite the good results, this approach is hungry for computational resources and time consuming, particularly in the model-tuning phase.

Sulaiman et al. [16] integrated graph neural networks (GNNs) with KANs. They introduced two new architectures for graph Kolmogorov–Arnold networks (GKANs), designed

to improve node classification in graphs. In the first architecture, they aggregated node features based on adjacency and then applied a KAN layer, whereas in the second architecture, they applied the KAN layer before aggregation. Both architectures demonstrated superior accuracy over traditional graph convolutional networks [17] on the Cora dataset [18]. The second architecture showed the highest performance, achieving more than 6% higher accuracy compared with the other models. The study also explored the effects of spline grid size, polynomial degree, and hidden layer size. Intermediate grid sizes and smaller polynomial degrees were better, and a hidden layer size of 12 proved effective, which may be because of the small size of the Cora dataset. These findings highlighted the potential of GKANs to advance graph representation learning, suggesting that they could replace conventional MLPs in various graph-based models. Li et al. proposed a framework integrating KANs with U-Net [19]. The U-KAN framework is a major advancement in medical image segmentation and generation by integrating KANs into the well-known U-Net architecture. They used a multilayered deep encoder–decoder structure with tokenized KAN blocks, validated on stringent medical segmentation benchmarks, such as breast ultrasound images [20], the Gland Segmentation in Colon Histology Images Challenge [21], and the CVC-ClinicDB database of colonoscopy videos [22]. However, the integration of KAN layers added complexity to the architecture, leading to longer training times and a dependency on high-quality training data.

IV. METHODOLOGY

A. KAN-SchNet

When KANs were introduced, we explored how they integrate with GNNs, so we began our formulation by modifying (13) by replacing $W(e_{i,j}) \odot x_j^l$ with

$$\sum_{i=0}^N c_i B_i(x) \quad (15)$$

Although $B_i(x)$ denotes the B-spline basis functions of degree N calculated with (6), and c_i is the learnable coefficients of the model, which resembles W in MLPs. One layer of KAN-SchNet can be defined as

$$x_k^{l+1} = x_k^l + \sum_{i=0}^N c_i B_i(e_{i,j}) \odot x_j^l \quad (16)$$

From a theoretical perspective, our new architecture could offer improved expressiveness and representational power, but the accuracy and efficiency need to be addressed. For example, the model could learn richer, more complex representations of molecular interactions with fewer parameters, making it more robust, although there are some drawbacks in terms of computational efficiency for KANs.

B. Cheby-KANs

The motivation behind modifying the B-spline KAN model was the huge overhead costs in terms of time and resources.

Initially, we planned to conduct more experiments in order to make a greater number of fair and robust comparisons, but the experiments consumed more time and resources than expected. Despite the remarkable results of KANs according to our experiments, it took 2 months to run these experiments on a supercomputer for approximately 130,000 molecular graphs of sizes up to 18 atom per molecule of QM9 dataset [23], [24], and that is not feasible in many cases. QM9 contains quantum chemical properties calculated at the DFT level for 134,000 small organic molecules, each composed of up to 9 heavy atoms (C, N, O, F). Also, it includes 19 geometric, energetic, electronic, and thermodynamic properties for each molecule, such as atomization energy, dipole moment, and HOMO-LUMO gap. After careful analysis of the algorithm, we reduced the number of parameters required by the model by using another type of well-behaved polynomial that has nice algebraical and geometrical properties to substitute for the B-splines' properties. The Chebyshev polynomials were selected based on their nice properties as described in Section II-B. We modified (15) with

$$\sum_{i=0}^N c_i T_i(x) \quad (17)$$

where $T_i(x)$ is the i^{th} Chebyshev polynomial of the first kind, which can be recursively defined as (12) for degree N , and c_i is the coefficients learnable by the model resembling W of MLPs.

The KANs use B-splines to construct smooth, well-behaved curves that can model complex relationships in high-dimensional data; however, KANs must construct these curves over a predefined grid, which can be fixed or learnable. However, this is a major limitation because this requirement adds huge computational expense if the grid is updated during the training time and the number of parameters is ($inputsize * gridsize + polynomialdegree + 1 * outputsize$). However, if the grid is fixed, then the model is not adaptable. In our model, the number of parameters for one layer was denoted with just ($inputsize * polynomialdegree + 1 * outputsize$), which decreased the overhead costs and time dramatically and achieved higher accuracies and lower losses in all experiments. The results are shown later in Section V of this paper.

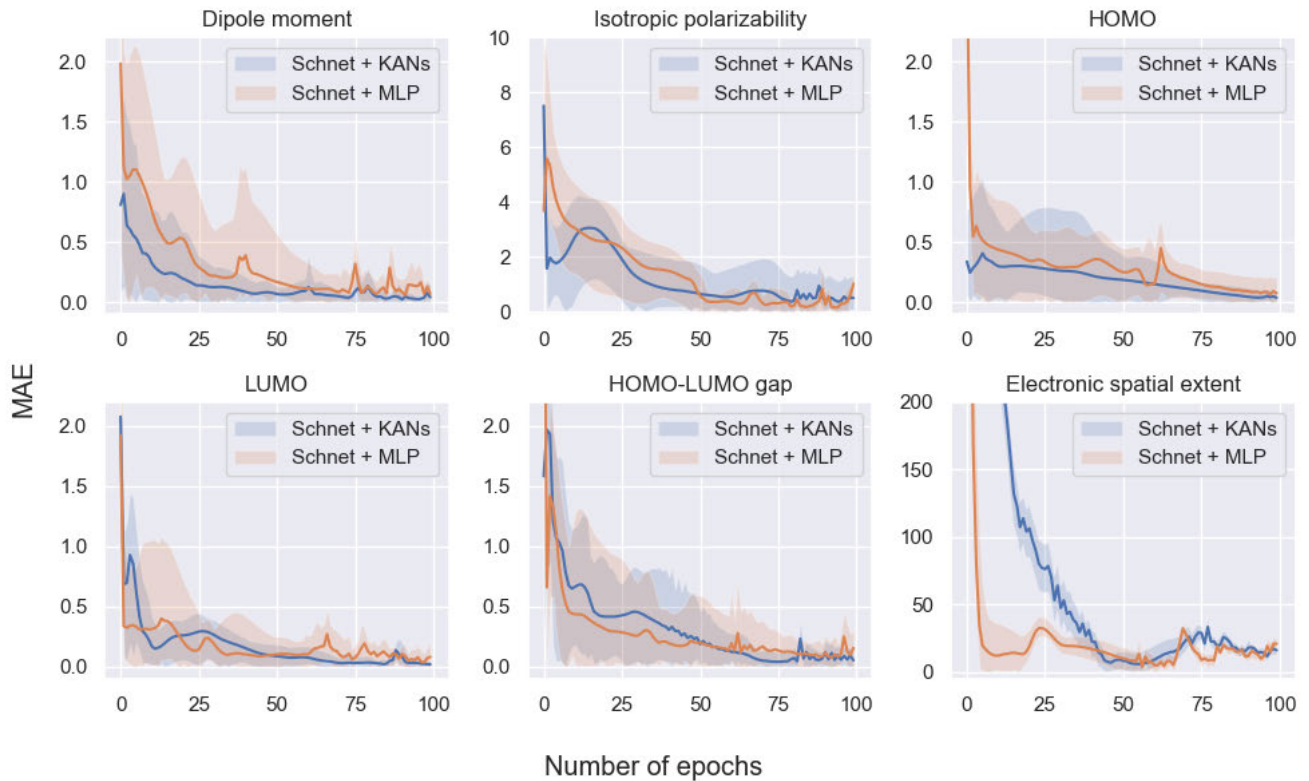
C. Cheby-KAN-SchNet

After validating our modified model described in Section IV-B on benchmark datasets, and after theoretical conceptual analysis, we were sufficiently confident to plug our model into the SchNet architecture. This was done by modifying (13) by replacing $W(e_{ij}) \odot x_j^l$ with (17), while keeping everything else the same. One layer of Cheby-KAN-SchNet is defined as

$$x_k^{l+1} = x_k^l + \sum_{i=0}^N c_i T_i(e_{i,j}) \odot x_j^l \quad (18)$$

TABLE 1. Comparison of KAN-SchNet and SchNet for various tasks. The table shows the MAE loss and a two-sample one-tailed t-test for the losses of the model with $H_0 : \mu_{\text{KAN-SchNet}} \geq \mu_{\text{SchNet}}$ and $H_1 : \mu_{\text{KAN-SchNet}} < \mu_{\text{SchNet}}$.

Task	KAN-SchNet	SchNet	Test Statistic	P-Value
Dipole moment	0.04 ± 0.03	0.07 ± 0.04	-23.3	1.27e-31
Isotropic polarizability	0.49 ± 0.35	1.0 ± 0.31	-17.42	1.71e-28
HOMO	0.03 ± 0.025	0.08 ± 0.06	-34.04	2.09e-56
LUMO	0.02 ± 0.02	0.08 ± 0.05	-16.63	1.67e-30
HOMO–LUMO gap	0.05 ± 0.04	0.16 ± 0.09	-35.23	1.26e-46
Electronic spatial extent	16.1 ± 1.6	21.0 ± 1.8	-23.93	6.12e-43

**FIGURE 1.** MAE loss calculation for each experiment. The plots indicating that KANs tend to be more consistent than MLPs in most experiments are based on the lower standard deviation (shaded areas).

This modification increased the expressiveness of the model and gave better results. After the modification, convergence speed was faster and stability was greater in comparison with B-spline KANs and SchNet. This improvement was attributed to the orthogonality of the Chebyshev polynomials helping to prevent the overfitting problem that the B-spline KAN is prone to by providing a compact, non-redundant representation of features. The smoother approximations offered by our choice of polynomials improved the model's ability to generalize for unseen data.

V. EXPERIMENTS AND RESULTS

We used the implementation of six regression tasks for predicting quantum properties using the QM9 dataset (Table 1). We did not alter the configurations provided by the PyTorch

Geometric (PYG) team [25] in their implementation for SchNet with regular MLPs, assuming that they set up the optimal configurations for the experiment, which is provided in the PYG documentation. In this section, we describe how the results were changed by introducing KANs into the architecture first with B-splines to evaluate the effect of KANs representation, and then we compare our enhanced Cheby-KAN model with the B-spline KANs when integrated with the SchNet model.

A. KAN- SchNet VS. SchNet

Our first experiment assessed the efficiency of SchNet by replacing the MLP parts with B-spline KANs. We expected KANs to be time- and resource-intensive, but during implementation we realized that it exceeded our expectations. The model achieved lower loss during training and evaluation

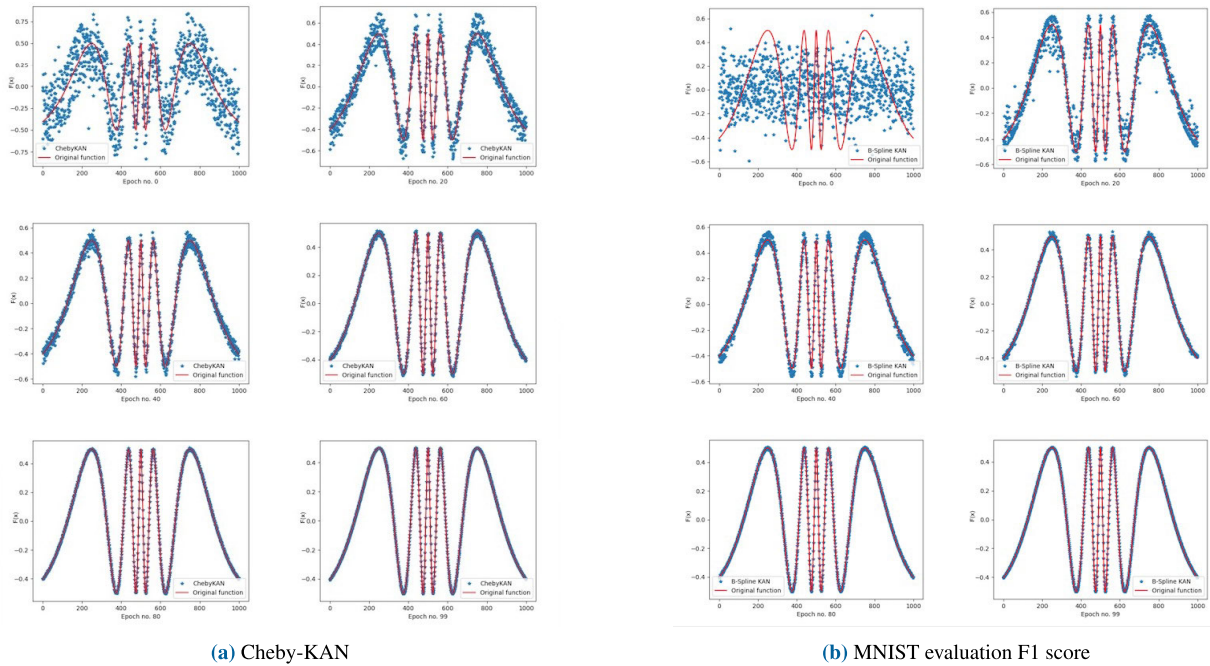


FIGURE 2. Performance of Cheby-KAN (a) and B-spline KAN (b) with a synthetic function. Convergence is faster with Cheby-KAN than with B-spline KAN. $F(x)$ is described by equation (19).

times, but with a huge difference in time; it took about 1 day for MLPs compared with 2 weeks for KANs. The results of this experiment are shown in Table 1 and Figure 1.

This experiment took more than 2 months on a supercomputer at Sorbonne University, Paris, France, and this was a big limitation because we needed to apply these models to larger datasets and different tasks. A practical requirement we found during training was the need to keep the learning rate low (a starting point of $1e-4$) and the momentum term of the optimizer low because KANs are more prone to instability when applied to deep architectures. This instability is caused by the additional matrix multiplications on the edges to estimate the non-linearities, leading to vanishing or exploding gradients. We argue that in general, smaller number of layers are needed for models that use KANs in their backbone than those that use MLPs. Nevertheless, this instability occurred only in the first learning batch, and when we scaled to the initial parameters, the model learned more efficiently than conventional SchNet. Moreover, the model showed greater convergence stability than the original SchNet, as the KAN representation decomposes the complex multivariate function into manageable single functions, reducing the overall complexity, minimizing potential sources of instability, and giving smoother optimization. This is indeed another advantage of KANs compared to MLP.

B. Cheby-KAN VS. ALL

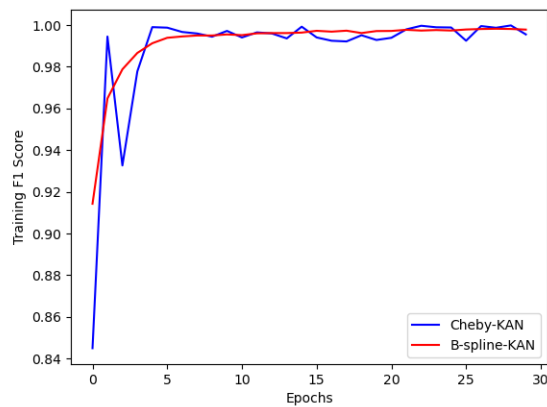
In this experiment, we tested our Cheby-KAN model against B-spline KANs and MLPs. The purpose of this experiment

was to evaluate our idea on benchmark datasets before integrating our model with the SchNet model. First, we tested our model on a dummy synthetic function to determine the time (2.3 times faster in this experiment compared with KAN-SchNet) and accuracy. The synthetic function tested the performance in regression tasks and the synthetic data generated by the function:

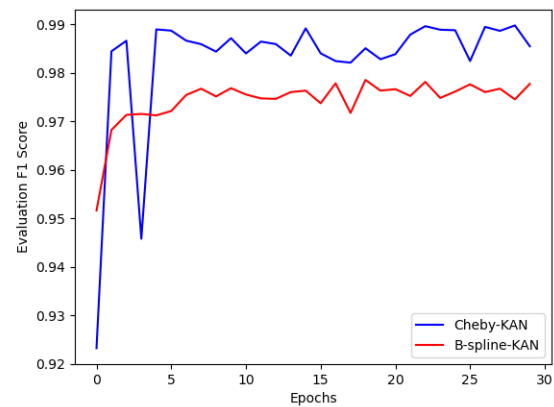
$$f(x) = 0.5 \cdot \cos\left(\frac{1.5 \cdot \pi \cdot 1.0}{|x| + 0.25}\right) \quad (19)$$

We were satisfied with these preliminary results because the model converged much faster towards learning the function in a much shorter time than the B-spline KANs, suggesting that our model was suitable. The results are shown in Figure 2.

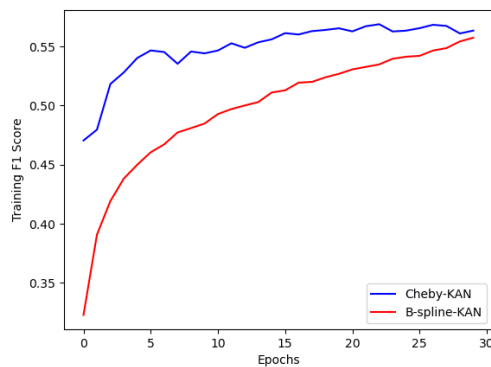
We used the same models for a multi-class classification task using the Iris dataset which contains 150 samples of iris flowers, with 4 features and three classes commonly used for classification tasks. The results are shown in Figure 4. Moreover, we have assessed the performance of our model (Cheby-KAN) on how it generalize to unseen data against the B-spline KAN. We have utilized two benchmark datasets (MNIST and CIFAR10) both are utilized from torchvision library. The results are shown in Figure 3. MNIST dataset consists of 70,000 grayscale images (60,000 for training and 10,000 for testing) of handwritten digits (0-9), each of size 28×28 pixels, widely used for benchmarking image classification models. On the other hand, CIFAR-10 dataset contains 60,000 color images (50,000 for training and 10,000 for testing) divided into 10 classes, such as airplanes, cars, and birds, with each image being 32×32 pixels, serving as



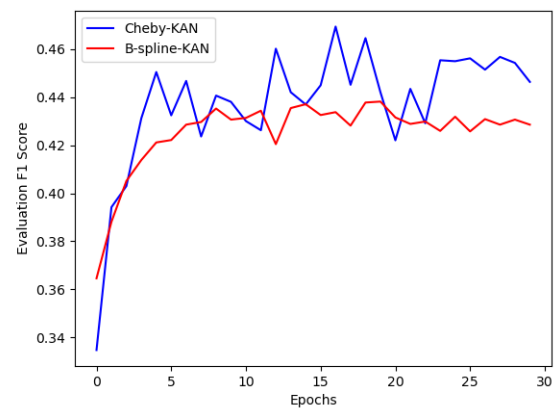
(a) MNIST training F1 score



(b) MNIST evaluation F1 score

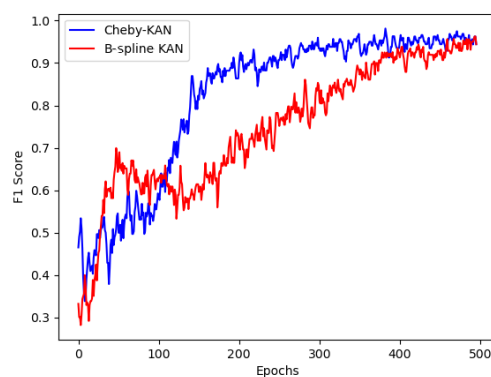


(c) CIFAR-10 training F1 score

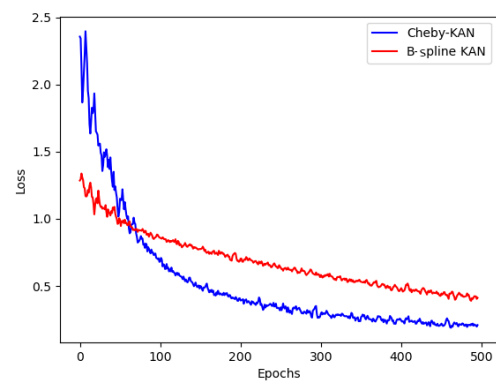


(d) CIFAR-10 evaluation F1 score

FIGURE 3. The efficiency of Cheby-KAN on unseen data outperforms the B-spline KAN on MNIST and CIFAR-10 datasets.



(a) F1 score



(b) Cross-entropy loss

FIGURE 4. Performance of Cheby-KAN and B-spline KAN on the multi-class classification Iris dataset. (a) F1 score and (b) cross-entropy loss. Cheby-KAN shows much faster convergence—achieving higher F1 scores and lower cross-entropy loss in fewer epochs—and is overall three times faster over the full 500 epochs.

a standard benchmark for more complex image recognition tasks. We have resized CIFAR10 to be of size (28×28) and transformed it into grayscale images. These datasets

are not related to quantum chemistry applications, yet they can provide insightful indicators for the performance of the models. Finally, While the curves do not exhibit full

TABLE 2. Comparison of MAE across different tasks for Cheby-KAN-SchNet, KAN-SchNet, and SchNet.

Task	Cheby-KAN-SchNet	KAN-SchNet	SchNet
Dipole moment	1.10	1.14	1.46
Isotropic polarizability	45.00	50.1	53.26
HOMO	2962.31	3483.55	3726.5
LUMO	1026.73	1099.63	2540.46
HOMO–LUMO gap	1330.53	4093.77	2204.05
Electronic spatial extent	575.39	581.02	578.88

TABLE 3. Comparison of time taken by each model.

Model	Time per iteration
SchNet	0.98 s
KAN-SchNet	14.5 s
Cheby-KAN-SchNet	1.10 s

convergence within the plotted epochs, this behavior aligns with our intentional design trade-offs, as we used early stopping and regularization mechanisms.

C. CHEBY-KAN-SchNet VS. ORDINARY SchNet, AND B-SPLINE KAN-SchNet

In this experiment, we compared the Cheby-KAN-SchNet model against the ordinary SchNet and B-spline KAN-SchNet models. The Cheby-KAN-SchNet model achieved higher accuracies, lower losses, and shorter time, and used fewer computational resources compared with the other model. We conducted this experiment on a smaller sample of the QM9 dataset consisting of 20,000 randomly selected molecules due to time and resources constraints, but we argue that this is still a valid comparison of models by induction. We followed the same procedures as described in Section V-A by predicting the same six properties. The results are shown in Tables 2 and 3. The Cheby-KAN-SchNet model was about 14 times faster than the B-spline KAN-SchNet model. Moreover, the mean absolute error (MAE) losses were lower for the Cheby-KAN-SchNet model than for the B-spline KAN-SchNet and original SchNet models.

D. INTERPRETATION AND APPLICABILITY OF OUR MODEL

The Chebyshev based KANs theoretically outperforms the B-spline KANs and MLP in the number of parameters needed for more accurate approximation due to its orthogonal polynomial bases, and this is also reflected from our results as it appears to converge faster with the same number of parameters. Moreover, unlike B-splines, which rely on piecewise polynomial interpolation and can suffer from local overfitting, Chebyshev polynomials offer a global approximation with exponential convergence for smooth functions, and this appears in the F1 score on the unseen data compared

to the B-spline based KANs. Additionally, Chebyshev KANs are computationally efficient due to the Clenshaw recurrence relation, which simplifies polynomial evaluation. It's worth mentioning that Cheby-KANs is much faster the B-spline KANs, but a little bit slower than MLPs, so it perfectly suits the scientific applications where a high accuracy is needed.

VI. CODE AVAILABILITY

All models, codes, and notebooks used in this article can be found for reproducibility at: IEEE data-port [26] and Zenodo [27]. All experiments including B-spline KANs were done using the efficient KAN implementation found in the following GitHub repository: <https://github.com/Blealtan/efficient-kan>. All experiments with MLP SchNet followed the implementation from the official PYG GitHub repository: https://github.com/pyg-team/pytorch_geometric.

VII. CONCLUSION AND FUTURE WORKS

In this work, we began to examine the impact of KAN representation on geometric DL because of the need for high accuracy when modeling scientific domains, particularly for quantum properties, owing to the exponential accumulation of errors. Our results show that KAN representation is promising for achieving high expressiveness and interpretability, and for learning complex and high-dimensional functions. However, the advantages of KANs come at the cost of longer times and higher computational costs. We considered mitigating some of the limitations by providing a faster, more efficient model by treating some gaps in the original model. We assessed our modified model objectively against benchmark datasets before putting it into a geometric DL model. Despite the encouraging results we achieved, we aim to improve our model further, for example, by tackling the problem of initial parameter scaling to make the model more compatible with deep architectures. It is hard to assess definitively whether Cheby-KANs or B-spline KANs will always outperform MLPs and this question will need years of practical experiments and theoretical interpretations of the results to make a robust decision; KANs are still in their infancy and require further practical examination.

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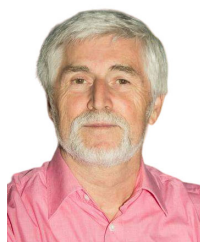
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AMGAD ABDALLAH MAHMOUD received the M.Sc. degree in data science from Cairo University, where he specialized in the integration of graph neural networks and deep reinforcement learning in generative AI for drug design. He is a Lecturer Assistant with The British University in Egypt (BUE), where he transfers knowledge and experience to his students. Moreover, he is dedicated to advancing pharmaceutical research by employing data-driven methodologies, including computational techniques and machine learning, to expedite the identification and development of new drugs. Also, his focus extends to biomedical informatics, where he explores the integration of mathematical modeling and domain knowledge for developing predictive models of treatment response and adverse reactions, and contributing to the advancement of therapeutic interventions and positively impacting individuals’ health. His research interests include but are not limited to developing reinforcement learning, graph neural networks, generative AI, and quantum machine learning algorithms to improve human health.



ANDREAS PESTER (Member, IEEE) is a Mathematician and a Data Scientist. He is a Professor with the Faculty of Informatics and Computer Science, BUE. He has more than 30 years of experience in teaching math and mathematical modeling, simulation technologies, online labs, and machine learning. Previously, he was responsible for the development of data science and deep learning research and teaching with the Carinthia University of Applied Sciences (Austria) and BUE (Egypt). He is also a reviewer of Ph.D. promotions with UPC Barcelona, UNED Madrid, and the University of Reading, U.K. He was a Guest Professor with UPC Barcelona, the Technical University of Kharkov, the Technical University of Kiev, University Maribor, UNESP Bauru (Brazil), the University of Applied Sciences Vienna, the Armenian–American University in Yerevan, PUC Rio de Janeiro, and University Banja Luka. In addition to his extensive scientific publications, he was involved in more than 20 EU and national projects in Austria and Egypt.



on mathematical modeling and analysis, contributing significantly to the body of knowledge in her area of expertise.



FREDERIC ANDRES (Senior Member, IEEE) received the Ph.D. degree in information systems from the University of Paris VI “Pierre et Marie Curie,” in 1993, and the Dr. Habilitation degree in informatics from the University of Nantes, in 2000. He has been an Associate Professor with the Digital Content and Media Sciences Research Division, National Institute of Informatics (NII), Tokyo, Japan, since 2000. He is the author of more than 150 publications for international journals, books, and conferences. His main research areas of interests include collective intelligence-based knowledge, AI ethics, data science, supply chain, and pedagogy and didactical web-based science education for all. His team won first prize in IEEE Brain Data Bank Challenges and Competitions at COMPSAC 2018, in July 2018. He received the IEEE CertifAIED Authorized Lead Assessor recognition from IEEE SA, in September 2023. In addition, he was named as a “R10 Ethics Champion” by IEEE Region 10, in October 2023. He has been co-organizing the International DECOR Workshop, since 2018, in conjunction with IEEE ICDE.



SHIHORI TANABE received the Ph.D. degree in pharmaceutical sciences from The University of Tokyo, Japan, in 2005. She is a Senior Researcher with the Division of Risk Assessment, Center for Biological Safety and Research, National Institute of Health Sciences (NIHS). Her doctoral research was in molecular biology and pharmacology. She has also carried out research focused mainly on biochemistry and cellular biology with the College of Medicine, University of Illinois at Chicago. A part of her research is focused on gene expression profiling and finding important factors of various cells, including cancer cells and stem cells using a bioinformatics approach including microarray and real-time RT-PCR techniques. She has also conducted research, such as cellular signaling, cardiac cell research, and protein regulation mechanism, and made contributions in the fields of molecular biology, pharmacology, and biochemistry. Her current research interest is analyzing molecular networks and profiling cancer cells and stem cells in terms of cellular phenotype and cellular signaling. She is also interested in finding essential factors to distinguish cancer cell types in order to reveal mechanisms of cancer development and the safe application of cells as cellular therapeutics in regenerative medicine, and in assessing the long-term influences of the alteration in gene expression and cellular phenotypes from the viewpoint of regulatory science. She has been a delegate of Japanese Society for Regenerative Medicine, a Councilor of the Society for Regulatory Science of Medical Products, a delegate of the Pharmaceutical Society of Japan, and an International Associate of Japanese Pharmacological Society.



teaching HPC and cloud infrastructure management and system and network security with the University of Sorbonne Paris North and ESIEE Paris.



HESHAM H. ALI is a Professor of computer science and the Director of the University of Nebraska at Omaha (UNO) Bioinformatics Core Facility. He was the Lee and Wilma Seemann Distinguished Dean of the College of Information Science and Technology, UNO, from 2006 to 2021. He has published several articles in various IT areas, including scheduling, distributed systems, data analytics, wireless networks, and bioinformatics. He has also published two books in scheduling and graph algorithms, and several book chapters in bioinformatics. He has been serving as the principal or co-principal investigator of several projects funded by NSF, NIH, and Nebraska Research Initiative, in the areas of data analytics, wireless networks, and bioinformatics. He has also been leading a research group that focuses on developing innovative computational approaches to model complex biomedical systems and analyze big bioinformatics data. The research group is currently developing several next generation big data analytics tools for analyzing large heterogeneous biological and health data associated with various biomedical research areas, particularly projects associated with infectious diseases, microbiome studies, early childhood development, and aging research. He has also been leading two projects for developing secure and energy-aware wireless infrastructure to address tracking and monitoring problems in medical environments, particularly to study mobility profiling for advancing personalized healthcare. He has led several local and national outreach initiatives, including Women in IT initiatives, IT education and training programs, and IT summer internship camps.

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